Ch 45 · Diagnosing acute kidney injury

KDIGO

- Creatinine ↑ ≥ 26.5 µmol/L in 48 h or ≥ 1.5× in 7 days, or UO < 0.5 mL/kg/h for 6 h
- Stage 1: 1.5–1.9×; stage 2: 2–2.9× / oliguria ≥ 12 h; stage 3: ≥ 3×, ≥ 354 µmol/L, RRT, anuria

Limits of creatinine

- Late rise, muscle mass, dilution, reduced production in sepsis

Newer tools

- NGAL, KIM-1, TIMP-2 × IGFBP7 (> 0.3); biomarker-guided bundle (PrevAKI)
- Furosemide stress test: < 200 mL in 2 h predicts progression
- Work-up: urine microscopy, ultrasound, review nephrotoxins

Key point: Urine output is the earliest KDIGO signal – measure it hourly.

Ch 46 · Managing AKI and RRT

General

- Restore perfusion, avoid overload, balanced crystalloids, stop nephrotoxins, dose drugs
- Diuretics for volume only

RRT indications

- Refractory K+, acidosis (pH < 7.15), overload with hypoxaemia, uraemic complications, toxins
- Timing: AKIKI, IDEAL-ICU, STARRT-AKI – no benefit from early start; AKIKI2 – excessive delay harmful

Modality

- CRRT if unstable or raised ICP; dose 20–25 mL/kg/h delivered (RENAL, ATN)
- Regional citrate preferred over heparin

Key point: Start RRT for indications, not for a creatinine number.

Ch 47 · Rhabdomyolysis

Mechanism of AKI

- Myoglobin casts, direct tubular toxicity, vasoconstriction, hypovolaemia

Causes

- Crush, immobilisation, exertion, seizures, hyperthermia, drugs (statins, cocaine), infection, low K/PO4

Management

- Treat hyperkalaemia first; aggressive crystalloid aiming UO 200–300 mL/h
- Bicarbonate/mannitol unproven; don't treat asymptomatic low Ca
- Check compartments (fasciotomy); RRT for refractory K+, acidosis, overload

Key point: Dipstick blood with no red cells = myoglobin.

Ch 48 · Hyponatraemia

Diagnosis

- Confirm hypotonic; urine osm < 100 = water excess; > 100 = ADH
- Urine Na > 30 + euvolaemia: SIADH, thiazide, hypothyroid, adrenal; < 30: hypovolaemia/oedema

Severe symptoms

- 3% saline 150 mL over 20 min, repeat ×2–3; aim +4–6 mmol/L
- SALSA: bolus as safe as infusion; check Na 2–4 hourly

Osmotic demyelination

- Limit ≤ 10 mmol/L/24 h (≤ 8 if alcohol, malnutrition, low K, liver disease)
- Overcorrection: D5W ± desmopressin to re-lower

Key point: Treat symptoms fast, then slow down – the first 5 mmol/L matters most.

Ch 49 · Hypernatraemia

Causes

- Water loss (intake, insensible, GI, osmotic diuresis), diabetes insipidus, sodium gain

Correction

- Free water deficit = TBW × (Na/140 − 1)
- Restore volume first if shocked; then water enterally or D5W plus ongoing losses
- Chronic: ≤ 10–12 mmol/L/day

Diabetes insipidus

- Polyuria with dilute urine and high Na; desmopressin response = central
- Central: desmopressin; nephrogenic: treat cause, thiazide

Key point: ICU hypernatraemia is often iatrogenic – watch saline and bicarbonate.

Ch 50 · Hyperkalaemia

ECG

- Peaked T → long PR, flat P → wide QRS → sine wave, VF/asystole

Emergency treatment

- Calcium chloride/gluconate until ECG improves
- Insulin 10 U + glucose; salbutamol 10–20 mg nebulised; bicarbonate if acidotic
- Remove K: stop ACEi/MRA, loop diuretic, binders (SZC, patiromer), dialysis

Causes

- Reduced excretion (AKI, drugs), shift (acidosis, insulin lack), cell breakdown, pseudohyperkalaemia

Key point: Calcium protects the heart but does not lower potassium.

Ch 51 · SIADH

Criteria

- Hypotonic hyponatraemia, urine osm > 100, urine Na > 30, euvolaemia
- Normal thyroid and adrenal function, no diuretics

Causes

- Small cell lung cancer, CNS disease, pneumonia/ventilation, drugs (SSRIs, carbamazepine), pain, nausea

Treatment

- Symptoms: 3% saline boluses
- Chronic: fluid restriction, oral urea, loop + salt; tolvaptan (overcorrection, liver risk)

Key point: Isotonic saline can worsen SIADH hyponatraemia.

Ch 52 · Diabetic ketoacidosis

Diagnosis

- Ketones ≥ 3, glucose > 11 (or known diabetes), pH < 7.3 or HCO3 < 15
- Severe: ketones > 6, HCO3 < 5, pH < 7.0, K < 3.5, GCS < 12

Treatment

- Fluids 1 L/h then tailored; FRIII 0.1 U/kg/h; continue basal insulin
- Add 10% dextrose when glucose < 14; K+ 40 mmol/L if 3.5–5.5; hold insulin if K < 3.5
- Targets: ketones ↓ 0.5/h, HCO3 ↑ 3/h, glucose ↓ 3/h

Complications

- Hypokalaemia, hypoglycaemia, cerebral oedema, hyperchloraemia, aspiration, VTE

Key point: Euglycaemic DKA occurs with SGLT2 inhibitors – check ketones.

Ch 53 · Thyroid storm

Diagnosis

- Clinical; Burch–Wartofsky ≥ 45; fever, AF, heart failure, CNS and GI-hepatic dysfunction
- Triggers: surgery, infection, contrast, stopping drugs, childbirth

Treatment order

- Beta blocker (propranolol/esmolol)
- PTU (or carbimazole) to block synthesis; iodine ≥ 1 h later
- Hydrocortisone; cooling, paracetamol (not aspirin); treat trigger
- Refractory: plasmapheresis, thyroidectomy

Key point: Thionamide before iodine – otherwise iodine fuels hormone synthesis.

Ch 56 · Hyperosmolar hyperglycaemic state

Diagnosis

- Osmolality ≥ 320, glucose ≥ 30, severe dehydration, ketones < 3, pH > 7.3

Treatment

- 0.9% saline first; osmolality ↓ 3–8 mOsm/kg/h
- Insulin 0.05 U/kg/h only when glucose stops falling with fluids (or ketotic)
- Glucose ↓ ≤ 5 mmol/L/h; replace K; VTE prophylaxis; treat trigger

Complications

- Cerebral oedema, osmotic demyelination, thrombosis, AKI, foot ulcers; mortality > DKA

Key point: In HHS fluids do most of the work – insulin comes later and lower.

Ch 57 · Myxoedema coma

Features

- Reduced consciousness, hypothermia, bradycardia, hypotension, hypoventilation
- Hyponatraemia, hypoglycaemia, effusions; triggers: infection, cold, sedatives

Treatment

- Hydrocortisone before levothyroxine
- IV levothyroxine 200–400 µg load then 50–100 µg/day (lower in cardiac disease)
- Ventilate for hypercapnia, passive rewarming, glucose, careful Na correction

Key point: Thyroid hormone without steroid can precipitate adrenal crisis.